

LifeLink

Blood Donation and Emergency Blood Matching System

A Relational Database Management System (DBMS) Project

Project Report

LifeLink is an advanced database-driven management platform engineered to automate emergency blood resource allocation, donor-recipient compatibility matching, location-aware donor fallback searches, and concurrent inventory reservation control.

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Contents

1	Project Overview	2
1.1	Introduction	2
1.2	Primary Goal	2
2	Problem Statement & Motivation	2
2.1	The Challenge in Emergency Blood Allocation	2
2.2	Core Problem Formulation	3
3	Existing Approach vs. Proposed Solution	3
3.1	Existing Workflow Analysis	3
3.2	Proposed LifeLink Solution	3
4	Project Objectives	3
4.1	Primary Objective	3
4.2	Specific Technical Objectives	3
5	System Architecture & Workflow	4
6	Database Architecture & Entity Relationships	4
6.1	Core Entity Specifications	4
6.2	Entity-Relationship Overview	5
6.3	Relational Table Specifications	5
7	Blood Matching Criteria & Logic	5
8	Demonstrated DBMS Concepts & Transaction Safety	6
8.1	Core DBMS Concepts Demonstrated	6
8.2	Transaction Safety & Concurrency Control	6
8.2.1	Double Reservation Scenario	7
8.2.2	Pessimistic Locking Implementation	7
9	Expected System Output & Features	7
10	Future Enhancements & Conclusion	8
10.1	Future Scope	8
10.2	Conclusion	8

1 Project Overview

1.1 Introduction

LifeLink is a centralized, database-driven blood management and emergency matching system engineered specifically for high-urgency healthcare environments. During clinical emergencies, time-to-transfusion is a critical parameter directly linked to patient mortality. LifeLink bridges the operational gap between hospitals, centralized blood banks, and voluntary donor networks by automating resource tracking and dynamic matching.

The core platform manages the end-to-end lifecycle of blood products and donor interactions across major functional domains:

- **Blood Donors:** Profiles, biological compatibility data, geo-coordinates, availability flags, and donation eligibility metrics.
- **Blood Inventory:** Individual unit tracking with collection, processing, and strict expiration timestamps.
- **Hospitals & Health Centers:** Emergency access credentials, institutional contacts, and spatial location data.
- **Emergency Blood Requests:** Urgency-rated patient blood demand logs.
- **Donation History:** Comprehensive records of past donations and biological deferral periods.
- **Blood Reservations:** Temporary and permanent allocation locks on inventory units.
- **Donor Availability & Response Tracking:** Real-time dispatching and confirmation of nearby eligible donors.

1.2 Primary Goal

The fundamental objective of LifeLink is to **dramatically reduce the operational lead time** required to locate, verify, and secure biological-compatible blood units or eligible emergency donors during critical patient care scenarios. By integrating real-time geolocation tracking, exact blood group compatibility matrices, biological eligibility rules, and strict ACID transaction locks, LifeLink replaces fragmented, manual workflows with an automated, fail-safe database pipeline.

2 Problem Statement & Motivation

2.1 The Challenge in Emergency Blood Allocation

In acute emergency scenarios (e.g., trauma, surgical hemorrhage, obstetrical complications), medical centers face severe logistical hurdles when securing vital blood supplies. Current workflows in many regions rely on decentralized phone calls, manual ledger lookups, and uncoordinated emergency appeals.

Key operational challenges include:

1. **Complex Biological Compatibility Filtering:** Ensuring recipient blood groups receive strictly compatible antigen matches under extreme time pressure.
2. **Inventory Visibility Gaps:** Difficulty in verifying real-time stock levels across distributed blood banks.
3. **Donor Eligibility & Recency Verification:** Rapidly evaluating donor interval eligibility (e.g., 90-day mandatory window between whole blood donations) without exposing patient safety.
4. **Geographical Proximity Isolation:** Locating available donors or blood units within an acceptable transport distance/time radius.

5. **Concurrency Control & Race Conditions:** Preventing double-reservation of a single scarce blood unit when multiple hospitals issue simultaneous emergency queries.
6. **Auditing and Record Integrity:** Maintaining immutable historical traces of blood utilization and safety compliance.

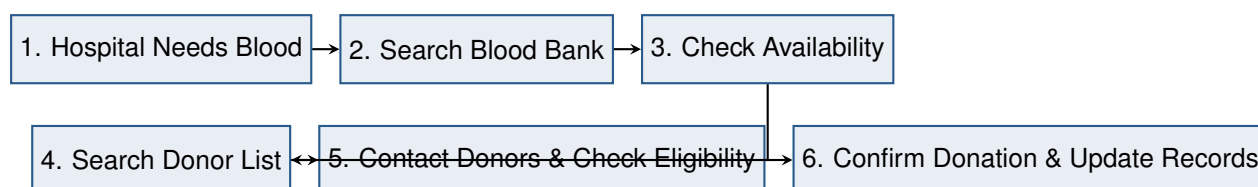
2.2 Core Problem Formulation

Core Question: *How can a relational database management system efficiently store, index, query, match, and lock blood resources during time-critical medical emergencies while ensuring absolute data integrity, zero race conditions, and optimized spatial proximity filtering?*

3 Existing Approach vs. Proposed Solution

3.1 Existing Workflow Analysis

The traditional manual approach suffers from latency, high error rates, and operational overhead. The sequential bottleneck is structured as follows:



Deficiencies of Existing Manual Systems:

- **Time-Consuming:** Takes hours to manually contact blood banks and individual donors.
- **Outdated Datasets:** Static donor lists frequently contain inactive numbers or ineligible individuals.
- **Lack of Coordination:** Uncoordinated appeals lead to donor fatigue and chaotic response handling.
- **Race Conditions:** High risk of duplicate reservations for a single scarce blood unit.
- **Incomplete Records:** Failure to maintain long-term historical tracking of donation cycles.

3.2 Proposed LifeLink Solution

LifeLink enforces an automated **Inventory-First with Donor-Fallback** workflow executed directly at the relational database engine level.

4 Project Objectives

4.1 Primary Objective

To design, implement, and validate a highly reliable, secure, and performant relational database system optimized for emergency blood matching and resource management.

4.2 Specific Technical Objectives

- **Centralized Donor & Facility Records:** Maintain unified schemas for donors, hospitals, inventory, and emergency dispatches.

Table 1: Comparison Between Existing Manual Approach and LifeLink

Feature	Traditional Manual Process	LifeLink Proposed DBMS
Matching Latency	Hours to days	Sub-second query execution
Compatibility Validation	Human error risk	Automated relational constraint logic
Inventory Control	Manual physical ledger checks	Real-time indexed database tracking
Donor Selection	Unfiltered sequential calls	Spatial distance & eligibility filtering
Concurrency Control	Prone to duplicate bookings	ACID-compliant transaction locking
Auditability	Partial/Fragmented paper logs	Immutable trigger-based audit logs

- **Real-Time Inventory Tracking:** Track unit lifecycles (Available, Reserved, Used, Expired).
- **Emergency Request Management:** Store and evaluate hospital emergency requests based on blood group and urgency levels.
- **Automated Compatibility Matching:** Execute SQL algorithms prioritizing biological antigen compatibility.
- **Eligible & Available Donor Identification:** Filter donors based on the mandatory deferral interval (90 days) and active status.
- **Location-Based Selection:** Support spatial proximity calculations to prioritize nearby available donors.
- **Concurrency Protection:** Deploy isolated transaction blocks (FOR UPDATE row locks) to eliminate duplicate blood reservations.
- **Data Integrity & Auditability:** Maintain complete historical logs and enforce foreign key constraints across all transactional entities.

5 System Architecture & Workflow

LifeLink operates on an inventory-first architecture. When a hospital creates a request, the system first attempts to lock stored physical inventory. If the required units are unavailable, the workflow automatically transitions to the secondary donor matching phase.

6 Database Architecture & Entity Relationships

6.1 Core Entity Specifications

The LifeLink relational schema is built around seven major core entities:

1. **DONOR:** Stores personal identification, ABO/Rh blood group, latitude/longitude coordinates, availability status, and last donation date.
2. **HOSPITAL:** Stores health center profile, emergency contact details, and location coordinates.
3. **BLOOD_UNIT / INVENTORY:** Tracks individual blood bags with collection date, expiry date, and status (Available, Reserved, Used, Expired).
4. **BLOOD_REQUEST:** Records emergency requests issued by hospitals, specifying target blood group, required units, and urgency level.

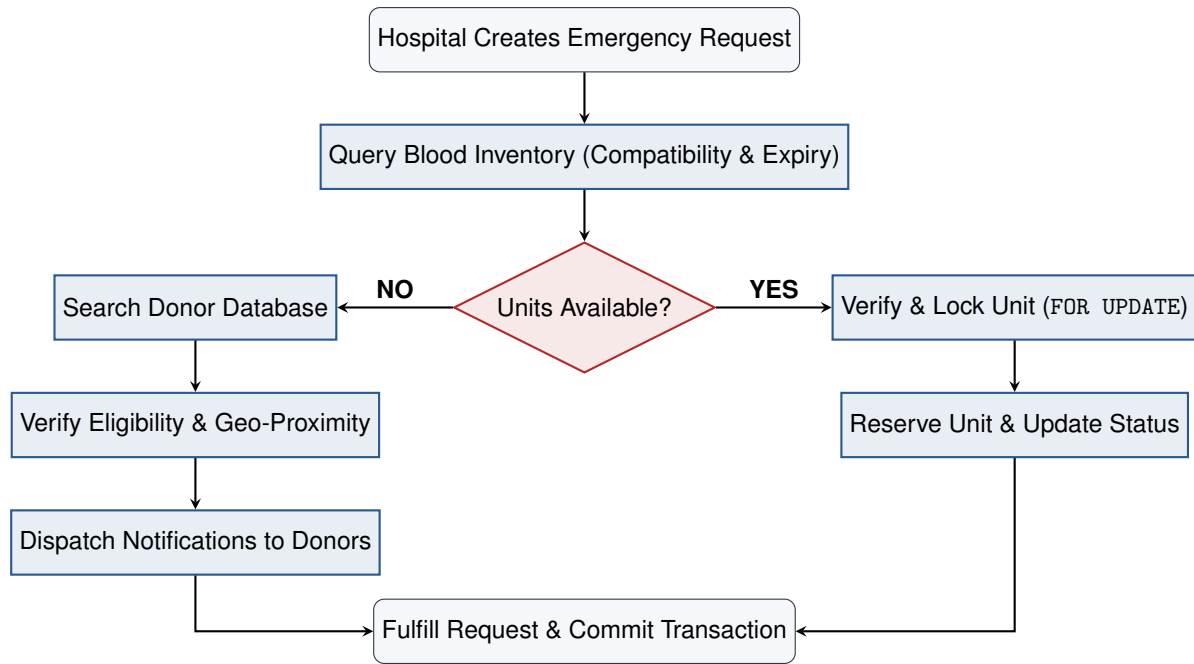


Figure 1: LifeLink System Workflow: Inventory-First and Donor-Fallback Process

5. **DONATION:** Historical log tracking specific donation events linking donors to produced blood units.
6. **RESERVATION:** Junction entity binding blood units or matched donors to specific blood requests.
7. **AUDIT_LOG:** Tracks critical system operations and state changes for auditing.

6.2 Entity-Relationship Overview

The primary relational dependencies among system entities are defined as follows:

$$\begin{aligned}
 &[\text{DONOR}] \xrightarrow{1 : N} [\text{DONATION}] \xrightarrow{1 : 1} [\text{BLOOD_UNIT}] \\
 &[\text{BLOOD_UNIT}] \xrightarrow{N : 1} [\text{RESERVATION}] \xleftarrow{1 : N} [\text{BLOOD_REQUEST}] \\
 &[\text{HOSPITAL}] \xrightarrow{1 : N} [\text{BLOOD_REQUEST}]
 \end{aligned}$$

Key Relational Rules:

- One donor can have multiple historical donations (1 : N).
- One donation produces specific blood inventory units (1 : 1).
- A hospital can create multiple emergency blood requests (1 : N).
- A blood request can link to one or more blood reservations (1 : N).
- Blood units dynamically transition through defined lifecycle states.

6.3 Relational Table Specifications

7 Blood Matching Criteria & Logic

The matching engine evaluates multi-attribute criteria before allocating a resource:

1. **Blood Compatibility Matrix:** Strict biological rules governing antigen-antibody interactions.

Table 2: DONOR Table Schema Specification

Attribute	Data Type	Constraint	Description
donor_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique donor identifier
name	VARCHAR(100)	NOT NULL	Donor full name
blood_group	VARCHAR(5)	NOT NULL	Blood group (e.g., O+, A-, AB+)
phone	VARCHAR(15)	UNIQUE, NOT NULL	Contact number
latitude	DECIMAL(10,8)	NOT NULL	Geographic latitude coordinate
longitude	DECIMAL(11,8)	NOT NULL	Geographic longitude coordinate
availability	BOOLEAN	DEFAULT TRUE	Current active donor availability
last_donation_date	DATE	NULLABLE	Date of most recent donation

Table 3: BLOOD_UNIT Table Schema Specification

Attribute	Data Type	Constraint	Description
unit_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique blood unit bag ID
blood_group	VARCHAR(5)	NOT NULL	ABO/Rh blood type
collection_date	DATETIME	NOT NULL	Timestamp of collection
expiry_date	DATETIME	NOT NULL	Expiration timestamp
status	VARCHAR(20)	CHECK (Status IN (...))	Status: Available, Reserved, Used, Expired

- *O- Negative*: Universal donor for all red blood cell transfusions.
- *AB+ Positive*: Universal recipient capable of receiving any ABO/Rh blood group.

2. **Availability Status**: Active validation that the donor/unit is marked as currently available.
3. **Donor Eligibility Rules**: Enforcement of safety deferral periods ($Current_Date - last_donation_date \geq 90$ days).
4. **Location & Proximity Filtering**: Dynamic ordering based on physical distance between donor coordinates and hospital location.
5. **Unit Expiry Validation**: Elimination of blood bags where $expiry_date \leq Current_Timestamp$.

8 Demonstrated DBMS Concepts & Transaction Safety

8.1 Core DBMS Concepts Demonstrated

- **Database Design**: ER modeling and schema mapping.
- **Data Integrity & Constraints**: Primary Keys, Foreign Keys, NOT NULL, UNIQUE, and CHECK constraints.
- **Normalization**: Adherence to Third Normal Form (3NF) to minimize update and deletion anomalies.
- **SQL Operations**: Complex JOIN queries, spatial sorting, subqueries, and aggregation.
- **Advanced DBMS Features**: Stored Procedures, Triggers, Indexes, Views, and ACID Transactions.

8.2 Transaction Safety & Concurrency Control

Preventing duplicate reservations under simultaneous multi-hospital queries is a core feature of LifeLink.

8.2.1 Double Reservation Scenario

Consider a situation where only one unit of O-negative blood (Unit #101) exists in inventory:

Hospital A → Requests Unit #101 — **Hospital B** → Requests Unit #101

Without concurrency protection, both requests would read Unit #101 as AVAILABLE, causing both to attempt reservation and corrupting inventory state.

8.2.2 Pessimistic Locking Implementation

LifeLink prevents this conflict using explicit row-level locking (FOR UPDATE) within a database transaction:

Listing 1: ACID-Compliant Reservation Transaction

```
BEGIN TRANSACTION;

-- Select and lock the eligible blood unit exclusively
SELECT unit_id FROM BLOOD_UNIT
WHERE blood_group = 'O-' AND status = 'AVAILABLE'
  AND expiry_date > CURRENT_TIMESTAMP
ORDER BY expiry_date ASC
LIMIT 1
FOR UPDATE;

-- Perform unit reservation
INSERT INTO RESERVATION (request_id, unit_id, reservation_time)
VALUES (301, 101, CURRENT_TIMESTAMP);

-- Update inventory status
UPDATE BLOOD_UNIT
SET status = 'RESERVED'
WHERE unit_id = 101;

COMMIT;
```

If any statement within the block fails, the database executes an automatic ROLLBACK, ensuring total data consistency.

9 Expected System Output & Features

The implemented system provides tailored interfaces for three major operational roles:

- **Hospital Interface:**

- Issue emergency requests with blood group and urgency indicators.
- View real-time matching units and eligible donors.
- Track request fulfillment status.

- **Donor Interface:**

- Register and update profile, blood group, and location.
- Toggle real-time availability.
- Receive emergency blood request notifications and respond (Accept/Reject).

- **System / Engine Functionality:**

- Dynamic inventory search and compatibility matching.

- Geographic proximity calculation and sorting.
- Automated row-level locking during reservation.
- Expiry management and audit logging.

10 Future Enhancements & Conclusion

10.1 Future Scope

- **Mobile Application Development:** Native iOS/Android app with instant push notifications.
- **Geographical Search Optimization:** Integration with mapping APIs (e.g., Google Maps API) for live traffic-aware routing.
- **Blood Demand Prediction:** Machine learning algorithms to predict regional demand spikes.
- **Hospital Inventory Sharing:** Inter-hospital blood exchange networks.
- **Automated Expiry Alerts:** Scheduled database jobs notifying blood bank staff of impending bag expirations.

10.2 Conclusion

LifeLink transforms blood management from a simple static record-keeping system into a structured, transaction-safe database solution for emergency blood matching. By enforcing biological compatibility rules, dynamic geo-proximity filtering, and ACID-compliant locking, LifeLink effectively resolves the core challenges of blood matching during critical emergencies.